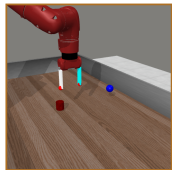
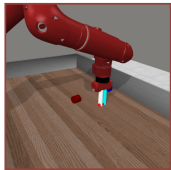


(a) Motivation



object shifted



ACT timeout

ACT keeps the stale action chunk



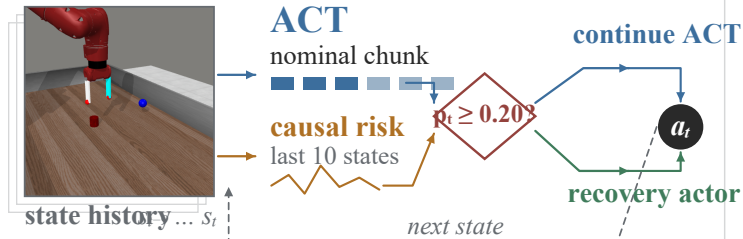
disturbance

state drifts away

outside demonstrated states

no recovery branch

(b) REIM *risk-gated selective recovery*



Recovery learns at trigger states

42,386 train / 8,212 validation pairs



trigger state



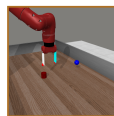
expert continuation

Smooth-L1
expert actions

π_{rec}

same visited states

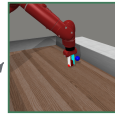
(c) Outcome



risk



re-grasp



success

Task success (%)

paired Δ +17.0 pp

95% CI +14.7 to +19.3

